

DEPARTMENT OF ELECTRONICS

Project Name: IoT based Smart Warehousing Robot Control with Mobile Application

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Branch: Robotics & Automation

Team Number: 4

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Project Abstract: The autonomous warehouse robot is designed for efficient movement using QR codes. It is controlled by a mobile app and can move forward, backward, or turn left and right, with each direction marked by color-coded icons. The robot's movement is powered by two DC motors controlled by a motor driver IC. To move forward, both motors are rotated clockwise, and to move backward, they are rotated counterclockwise. Left or right turns need one motor to turn clockwise and the other counterclockwise. A power-off mode is also available for easy shutdown. Sensors have been added for added safety and efficient navigation.

Smart warehouse robots have several key applications, including:

1. Automated Picking & Packing – Robots pick and pack items efficiently, reducing human effort and errors.
2. Inventory Management – AI-powered robots track and update stock levels in real time.
3. Sorting & Categorization – Automated sorting of products based on destination or category.
4. Material Handling – Transporting goods within the warehouse, improving logistics flow.
5. Order Fulfillment – Speeding up the processing and shipping of online orders.
6. Quality Control – Inspecting and ensuring product quality before dispatch.
7. Workforce Assistance – Collaborating with human workers to enhance productivity and safety.

